BANK INTEREST DEMO

Aim

To write a Java program that demonstrates method overriding and runtime polymorphism by displaying the rate of interest offered by different banks.

Description

The program defines an RBI class with a default interest rate. SBI, ICICI, and PNB extend RBI and override the getRateOfInterest() method with their own rates. The user enters a bank name, and the program creates the matching bank object and displays its interest rate. If the name is not recognized, an error message is displayed.

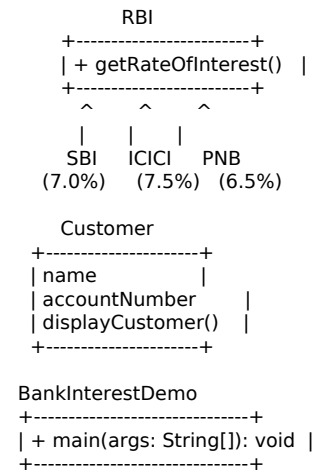
Software Requirements

Hardware: Computer or laptop. Software: Java Development Kit (JDK) and a Java IDE or text editor such as Notepad, VS Code, or Eclipse.

Algorithm

1. Start the program.
2. Create a Scanner object and read the bank name.
3. Remove extra spaces from the entered name.
4. Declare an RBI reference variable.
5. Create an RBI, SBI, ICICI, or PNB object according to the entered name.
6. If the bank name is invalid, display an error message and stop.
7. Display the bank name and the interest rate returned by getRateOfInterest().
8. Close the Scanner and stop.

UML Diagram



Code

```
package recordprograms;

import java.util.Scanner;

class RBI {
    double getRateOfInterest() {
        return 4.0;
    }
}

class SBI extends RBI {
    @Override
    double getRateOfInterest() {
        return 7.0;
    }
}

class ICICI extends RBI {
    @Override
    double getRateOfInterest() {
        return 7.5;
    }
}

class PNB extends RBI {
    @Override
    double getRateOfInterest() {
        return 6.5;
    }
}

class Customer {
    String name;
    String accountNumber;

    Customer(String name, String accountNumber) {
        this.name = name;
        this.accountNumber = accountNumber;
    }

    void displayCustomer() {
        System.out.println("Customer Name: " + name);
        System.out.println("Account Number: " + accountNumber);
    }
}

public class BankInterestDemo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter bank name: ");
        String name = sc.nextLine().trim();

        RBI bank;

        if (name.equalsIgnoreCase("RBI")) {
            bank = new RBI();
        }
        else if (name.equalsIgnoreCase("SBI")) {
            bank = new SBI();
        }
        else if (name.equalsIgnoreCase("ICICI")) {
            bank = new ICICI();
        }
        else if (name.equalsIgnoreCase("PNB")) {
            bank = new PNB();
        }
        else {
            System.out.println("Invalid bank name!");
            sc.close();
            return;
        }

        System.out.println("Bank Name: " + name.toUpperCase());
        System.out.println("Rate of Interest: "
                + bank.getRateOfInterest() + "%");

        sc.close();
    }
}
```

Output Screenshot

Example 1: SBI

```
Enter bank name: SBI
Bank Name: SBI
Rate of Interest: 7.0%
```

Example 2: ICICI

```
Enter bank name: ICICI
Bank Name: ICICI
Rate of Interest: 7.5%
```

Result

Thus, the Java program demonstrating method overriding and runtime polymorphism to display different banks' rates of interest was written and executed successfully.